MIL-P-13850C
1 March 1970
SUPERSEDING
MIL-P-13850B
30 March 1964

MILITARY SPECIFICATION

PANEL MARKER, AERIAL LIAISON, COTTON SHEETING, PANEL AL-121, WHITE AND PANEL AL-122, BLACK

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope.- This specification covers one type and size of panel marker in two colors. Panel marker AL-121 is bleached white and panel marker AL-122 is dyed black.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-375 - Plastic Film, Flexible, Vinyle, Chloride
V-T-285 - Thread, Polyester
UU-P-268 - Paper, Kraft, Untreated, Wrapping
CCC-C-432 - Cloth, Sheeting, Cotton (Unbleached, Bleached, and Dyed)
CCC-D-950 - Dyeing and Aftertreatment Processes for Cotton Cloths
DDD-L-20 - Label; for Clothing, Equipage, and Tentage (general Use)
PPP-B-601 - Boxes, Wood Cleated, Plywood
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner
PPP-B-636 - Box, Fiberboard
PPP-T-45 - Tape, Gummed, Reinforced and Plain for Sealing and Securing

MILITARY

MIL-G-16491 - Grommets, Metallic

/FSC 8345/

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STANDARDS

FEDERAL

Fed. Std. No. 191 - Textile, Test Methods
Fed. Std. No. 751 - Stitches; Seams; and Stitchings

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage

DRAWINGS

THE INSTITUTE OF HERALDRY

5-1-330 - Panel Marker, Aerial Liaison, Cotton Sheeting; Panel AL-121, Color White or Panel AL-122, Color Black; Assembly

(A miniature copy of drawing 5-1-330 identified as figure 1 is attached for information only.)

(Copies of specifications, standards, and drawings required by the contractor in connection with a specific procurement function should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

HANDBOOK OF COLORIMETRY (MIT PRESS)

(Application for copies should be addressed to the Technology Press; Massachusetts Institute of Technology; Cambridge, Massachusetts 02138.)

3. REQUIREMENTS

3.1 First article.- Unless otherwise specified (see 6.2), before production is commenced a sample of the finished panel shall be submitted or made available to the contracting officer or his authorized representative for inspection and tests as specified in 4.2. The approval of the first article

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authorizes the commencement of production but does not relieve the contractor of responsibility for compliance with all provisions of this specification. The first article shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Materials.- Materials shall conform to the referenced specifications and to the requirements specified herein.

3.2.1 Cloth, cotton sheeting.- The cotton sheeting shall conform to type VI, class 2 of CCC-C-432, except that only the following characteristics shall apply: threads per inch, weight, breaking strength, shrinkage, and sulphur content. The finished sheeting shall be bleached for panel AL-121 and dyed for panel AL-122.

3.2.1.1 Color.-

3.2.1.1.1 Panel AL-121.- The panel shall be bleached white conforming to the shade of the standard (see 6.3). Bleaching of the fabric by any commercial method is permissible. The use of a fluorescent brightener is required. When tested as specified in 4.4.1, the finished panel shall have a luminous reflectance relative to magnesium oxide of not less than 75.0 percent. When tested as specified in 4.4.1 and 4.4.2, the brightener shall be fast to chlorine bleaching and shall have a spectral emission with a peak response in the blue of the spectrum. When tested as specified in 4.4.3, the pH of the finished fabric shall be not less than 5.0 nor more than 8.0. When tested as specified in 4.4.4, after 20 hours exposure, the finished fabric shall show no more yellowness than the standard or when the standard is not available the colorfastness to light shall be good.

3.2.1.1.2 Panel AL-122.- The panel shall be dyed black conforming to the shade of the standard (see 6.3). The black shall be obtained by the use of anthraquinone vat dyes. The use of sulfur dyes or aniline black is prohibited. When tested as specified in 4.4.1, the finished panel shall have a luminous reflectance relative to magnesium oxide of less than 3.0 percent and shall have an integrated visual reflectance with regard to the standard observer and CIE luminat C of less than 4.0 percent. When tested as specified in 4.4.4, after 100 standard fading hours, the color fastness of the dyed fabric shall be equal to or better than the standard sample. When a standard is not established or not designated as applicable to colorfastness, the dyed fabric shall show good fastness to light.

3.2.1.2 Mildew-resistant treatment.- The cotton sheeting shall be mildew-resistant treated in accordance with table III of CCC-D-950. The bleached sheeting shall be treated with inhibiter (i) and the dyed sheeting with inhibitor (a).

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3.2.2 Thread, polyester.- Thread shall conform to class 1, type I or II, size B of V-T-285, except that only the following characteristics shall apply; identification of polyester, breaking strength, elongation, and colorfastness to light (20 hours). The characteristics on colorfastness to light shall apply to the panel marker AL-122 only.

3.2.2.1 Color.- The thread used in sewing the panel AL-121 shall be natural in color. The thread used in sewing the panel AL-122 shall be black and shall show fastness to chlorine bleaching equal to or better than the standard (see 6.3). When standard is not available, the thread shall show good fastness to chlorine bleaching.

3.2.3 Plastic washers.- The plastic washers shall be in accordance with the requirements of drawing 5-1-330 and shall be fabricated from plastic conforming to L-P-375 except only the following characteristics shall apply: Resistance to weathering, cold crack at minus 40 degrees C ± 2 degrees C., and dimensional stability. The washers for panel AL-121 shall be formed using type II, class 1 plastic and the washers for panel AL-122 shall be formed using type II, class 2 plastic.

3.2.4 Grommets.- The grommets shall conform to type II, class 3, size No. 2-J of MIL-G-16491, except only the following characteristics shall apply: chemical composition, dimensions and finish.

3.3 Construction.- The panel markers shall be constructed in accordance with drawing 5-1-330 and as specified herein.

3.3.1 Stitching.- All stitching shall conform to type 301 of Std. No. 751, 6 to 8 stitches per inch. Stitching shall be back stitched at the ends not less than 1/2 inch, except where ends are turned under in a hem and held down by other stitching. Thread breaks in stitching shall be back stitched not less than 1 inch at each break. Thread tension shall be maintained so that there will be no loose stitching, and that the lock will be embedded in the center of the material sewed. All threads ends shall be trimmed.

3.3.2 Splicing of panels.- One splice per panel will be allowed. Seams shall conform to type LSc-2 and shall be 1/4 to 5/16 inch gauge. Type 301 stitching shall be used. The one splice allowed shall be in the short dimension of the panel marker.

3.3.3 Setting of grommets.- Holes punched to receive grommets specified in 3.2.4, shall be smaller than the outside diameter of the grommet barrel so that the barrel must be forced through the hole. The grommets shall be securely clinched without cutting the material.

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3.4 Marking.- The letters "U.S." and the identification marks shall be printed or stenciled in the size characters and in the locations as shown on drawing 5-1-330. The AL-121 panel shall be marked with black ink and the AL-122 panel shall be marked with white ink. The identification marking shall conform to the requirements for type IV, class 5 labels of DDD-L-20. Fastness shall be as specified for type IV, class 5 labels. All marking shall be on the hem side.

3.5 Workmanship.- Workmanship for the panels shall conform to the quality and grade product established by this specification and the occurrence of defects shall not exceed the acceptable quality levels.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection responsibility.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requiremen

4.2 First article inspection.- Inspection and testing of the first article shall be made on a completely fabricated panel marker for all provisions of this specification applicable to end product examination and test.

4.3 Sampling for inspection and acceptance.- Sampling for inspection and acceptance shall be in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated.

4.3.1 Testing of components.- In addition to the quality assurance provisions of the subsidiary specifications and drawings, testing shall be performed on components and materials listed in table I for the characteristics specified therein. Where applicable the inspection level shall be S-1 and the acceptable quality level (AQL) shall be 4.0. All test methods cited are from Std. No. 191.

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4.3.1.1 Certification of compliance.- The cotton sheeting and the chemical composition of the grommets may be accepted on the basis of a contractor's certification of compliance.

4.3.2 In-process inspection.- Inspection shall be made at any point or during any phase of the manufacturing process to determine whether operations or assemblies are accomplished as specified. The Government reserves the right to exclude from consideration for acceptance any material or service for which in-process inspection has indicated nonconformance. In-process inspection shall be conducted to see that accomplishment of the following is in accordance with the specification requirement:

Requirement, operation or assembly	Requirement paragraph
Holes punched for grommets are smaller than outside diameter of grommet barrel.....	3.3.3

4.3.3 Inspection of the end item.-

4.3.3.1 Visual examination of panel marker, aerial liaison, cotton sheeting.- The defects found during examination will be classified in accordance with 4.3.3.1.1 and 4.3.3.1.2.

4.3.3.1.1 Examination of panel marker, aerial liaison, cotton sheeting for defects in material, construction, and workmanship.-

Examine	Defect	Classification	
		Major	Minor
Quality of material	Any hole, cut or tear.....	x	-
	Any smash.....	x	-
	Floats or skips:		
	-multiple, 1/2 inch or more in combined directions.....	x	-
	-multiple, but less than 1/2 inch in combined directions.....	-	x
	-single float or skip over 12 inches in length.....	-	x
	Broken or missing ends:		
	-three or more contiguous regardless of length.....	x	-
	-two contiguous more than 3 inches in length.....	-	x
	-singles, less than 6 inches apart.....	-	x
	-singles, more than 36 inches in length....	-	x
	-singles, more than three per panel up to and including 36 inches in length.....	-	x

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Examine	Defect	Classification	
		Major	Minor
	Broken picks:		
	-three or more contiguous regardless of length	x	-
	-two contiguous regardless of length	-	x
	-singles, less than 6 inches apart	-	x
	Mispick or double pick:		
	-two or more contiguous	-	x
	-singles, less than 6 inches apart	-	x
	Slack end resulting in loops, curls or kinks on the surface of the fabric extending more than 1/2 inch in length	-	x
	Any fine or light filling bar or light place clearly noticeable	-	x
	Open place (poor scratch up):		
	-open between yarns 1/16 inch or more at widest place	x	-
	-opening between yarns less than 1/16 inch resulting in distorted texture over 1/2 inch square in area	-	x
	-more than two, 1/2 inch square or less in area, within a 1 foot square area	-	x
	Slub, slug, gout or sloughoff:		
	-1/8 inch or more in thickness	x	-
	-more than 1/16 inch but less than 1/8 inch in thickness	-	x
	Spot or stain, 1/8 inch or more clearly noticeable when viewed at a distance of 6 feet	-	x
	Wrong draw, two or more extending 36 inches or more in length	-	x
Grommets, metallic	Finish not type specified or evidence of corrosion	-	x
	Sharp edge or projection that may cause injury	x	-
	Any missing	x	-
	Insecurely clinched, broken, cracked, or otherwise defective, seriously affecting serviceability	-	x
	Insecurely clinched, broken, cracked or defective, affecting serviceability, but not seriously	-	x

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Examine	Defect	Classification	
		Major	Minor
	Clinched too tight, cutting surrounding fabric	x	-
Plastic washer	Any missing	x	-
	Wrong color	-	x
	Broken or cracked affecting seal	-	x
Workmanship of seam and stitching	Incorrect hem or stitch type	x	-
	One or two stitches per inch less than minimum specified	-	x
	Less than 4 stitches per inch	x	-
	One or more stitches in excess of maximum specified	-	x
	Thread breaks back stitched less than one inch	-	x
	End of stitching back stitched less than 1/2 inch except where ends are turned under or caught in other stitching	-	x
	Needle chews or excessive number of stitches per inch which results in cutting of fabric	x	-
	Back stitching omitted where required	-	x
	Tight tension, resulting in excessive puckering	-	x
	Loose tension	-	x
	Thread end not trimmed	-	x
	Thread not color specified	-	x
	Open seam up to and including one inch in length	-	x
	Open seam more than 1 inch long	x	-
	Raw edge up to and including 2 inches in length	-	x
	Raw edge more than 2 inches long	x	-
	Stitching margin or stitch gauge more or less than specified	-	x
	Two contiguous skip stitches	-	x
	Two or more skips in two inches of seam	-	x
	NOTE: More than two skipped stitches shall be classified an open seam.		
	Run off up to and including one inch in length	-	x
	Run off over one inch	x	-

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Examine	Defect	Classification	
		Major	Minor
Splicing of panels	More than one splice per panel.....	x	-
	Stitching gauge less than 1/4 inch or more than 5/16 inch.....	-	x
	Splice other than in short dimension.....	-	x
	Stitch type other than 301.....	-	x
	Seam type other than LSc-2.....	-	x
Marking	Omitted, illegible or incorrect.....	-	x
	Located other than as shown on drawing....	-	x
	Color not as specified.....	-	x
	Characters less than 15/16 inch or more than 1-1/16 inch.....	-	x

4.3.3.1.2 Examination of panel marker, aerial liaison, cotton sheeting, dimensions.- Any dimension that exceeds the specified tolerances shall be classified as a defect.

4.3.3.1.3 Inspection levels and acceptable quality levels (AQL's).- The inspection levels and acceptable quality levels (AQL's) expressed in defects per hundred units shall be as follows:

	Inspection level	AQL Major	AQL Total
For examinations applicable to 4.3.3.1.1	II	4.0	10.0
For examinations applicable to 4.3.3.1.2	S-3	(one class)	10.0

4.3.4 Examination of preparation for delivery requirements.- An examination shall be made to determine that the packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with the list below. The sample unit shall be one shipping container fully packed, selected just prior to the closing operation. Defects of closure listed below shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level shall be S-2 and the AQL shall be 2.5 defects per hundred units.

Examine	Defect
Marking (exterior and interior)	Omitted, incorrect, illegible, or improper size, location, sequence or method of application.
Materials	Any component missing.
	Any component damaged, affecting serviceability.
Workmanship	Inadequate application of components, e.g., incomplete closure of case liners, container flaps, loose strapping, inadequate stapling.
	Bulging or distortion of containers.
Content	Number per container is more or less than required.

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4.4 Tests.-

4.4.1 Colorimetry.- Data shall be determined in accordance with procedures outlined in Handbook of Colorimetry (MIT Press), and rules covered by the International Committee on Illumination, based on Spectral Evaluation by Beckman or GE Spectrophometry.

4.4.2 Test for chlorine fastness.- The chlorine fastness of the fluorescent brightener shall be equal to the standard sample when tested in accordance with method 5600 of Std No. 191. The fluorescent emission shall be checked under an ultraviolet source having a spectral distribution covering the wavelengths 0.3 microns to 0.4 microns with a maxima at approximately 0.36 microns.

4.4.3 Test for pH.- The pH of the finished fabric shall be determined in accordance with method 2811 of Std. No. 191.

4.4.4 Colorfastness of fabric.- Samples shall be exposed in accordance with method 5660 of Std. No. 191.

5. PREPARATION FOR DELIVERY

5.1 Packaging.- Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.- Each panel shall be neatly and smoothly folded to approximate dimensions of 18 inches by 7 inches. Five folded panels, of one color only, shall be neatly and evenly stacked in a bundle and securely tied approximately four inches from each end with cotton tape or twine. Each bundle shall be completely wrapped in 30-pound minimum basis weight kraft paper conforming to grade B of UU-P-268. Secure the kraft paper wrap with 2-inch minimum with gummed paper tape conforming to PPP-T-45.

5.1.2 Level C.- Panels shall be packaged to afford protection from deterioration and damage during shipment from the supply source to the first receiving station. The supplier may use his commercial practice when it meets these requirements.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Nine bundles (45 panels) of one color only, packaged as specified in 5.1, shall be packed flat, three in length, one in width and three deep, in a snug fitting shipping container conforming to class 2 of PPP-B-621 or overseas type, style A, B, I or J of PPP-B-601. Closure shall be in accordance with the appendix of the applicable container specification.

5.2.2 Level B.- Nine bundles (45 panels) of one color only, packaged as specified in 5.1 shall be packed flat, three in length, one in width and three deep, in a snug fitting fiberboard shipping container conforming to class

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domestic, grade 275 of PPP-B-636. Closure shall be in accordance with the appendix of the box specification.

5.2.3 Level C.- Panels of one color only, packaged as specified in 5.1, shall be packed in a manner to assure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with the rules and regulations of carriers applicable to the mode of transportation.

5.3 Marking.- Interior packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use.- The panels covered by this specification are intended to be used in ground to air communications and for identification of ground units by aircraft personnel.

6.2 Ordering data.- Procurement documents should specify the following:
(a) Title, number and date of this specification.
(b) Panel marker required (see 3.2.1.1).
(c) When a preproduction sample is not required (see 3.1).
(d) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.3 Standard shade sample.- For access to a standard shade sample of cloth, a standard sample of thread, and markings address the procuring officer issuing the invitation for bids or request for proposal.

Custodians:

Army - IH
Air Force - 82

Review Activities:

Air Force - 82
Other - DP

User Activities:

Army - SM
Navy - MC

Preparing Activity:

The Institute of Heraldry
United States Army

Project No. 8345-0072

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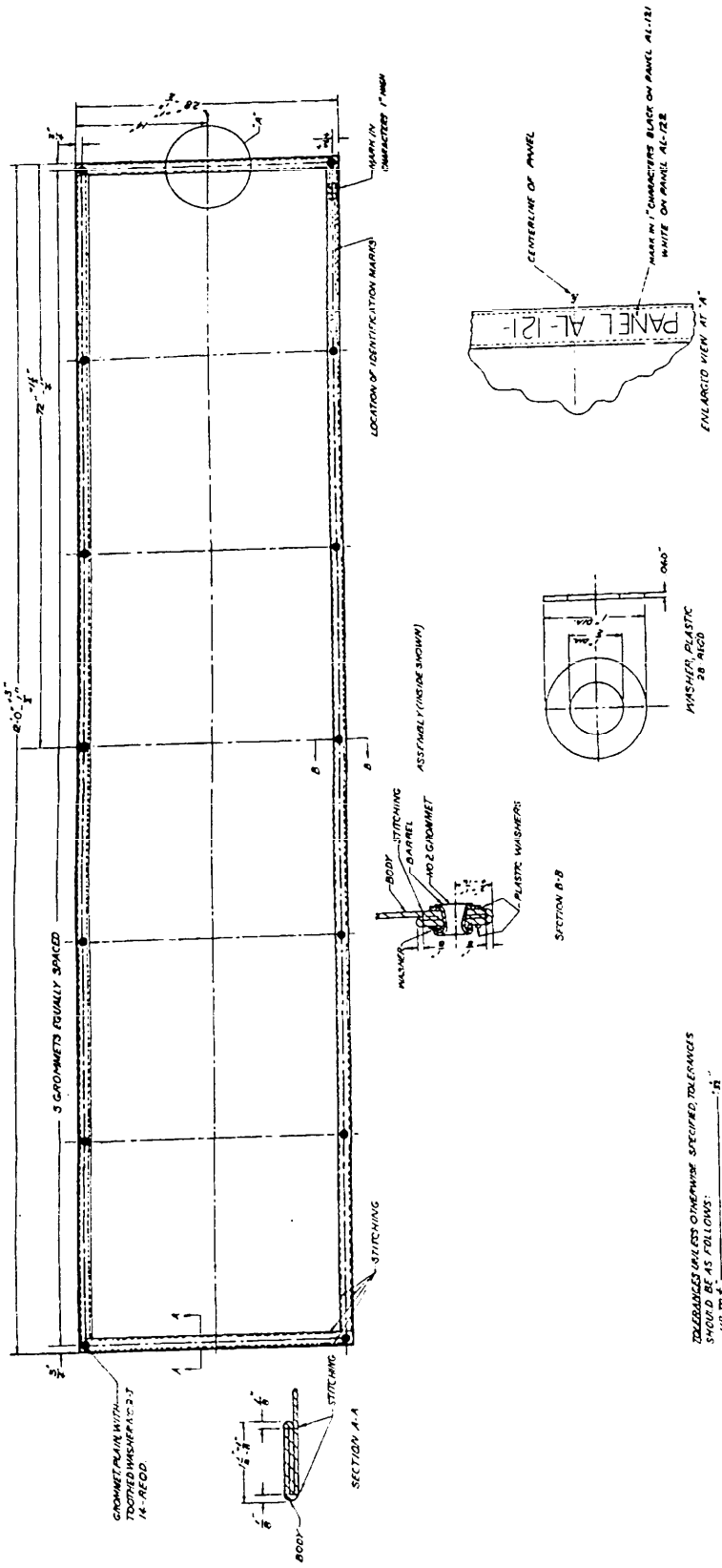


FIGURE 1

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION MIL-P-13850C Panel Marker, Aerial Liaison, Cotton Sheeting, Panel AL-121, White and Panel AL-122, Black

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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